

# Design-analysis Question Bank (Identify violations + justify)

## Q1) “Hidden voucher” in a ride-booking app

**Scenario:** On the checkout screen, “Apply Voucher” is hidden inside a tiny “More” icon. After applying, there is no visible confirmation; the price looks the same until you scroll.

**Model answer:**

This violates **Recognition rather than recall** because users must *remember* that vouchers exist and where they are hidden instead of *recognizing* the option directly on the checkout UI. It also violates **Visibility of system status** because the system does not clearly show that the voucher was applied (no immediate status/price change feedback).

From Shneiderman, it violates **Offer informative feedback** and **Design dialogs to yield closure** because after an action (“apply voucher”), users don’t get a clear “done + updated total” endpoint.

Result: users think the feature is broken, lose trust, and may abandon checkout. Fix: make “Apply Voucher” **visible on the main checkout**, show an **immediate total update**, and add explicit confirmation like “Voucher applied: -Rs100”.

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## Q2) “No undo” when selecting pickup time

**Scenario:** User selects a wrong pickup time. There is no cancel/undo and the only way is to fully restart the booking.

**Model answer:**

This violates **User control and freedom** because the user cannot easily exit or undo a wrong action. It also violates Shneiderman’s **Permit easy reversal of actions**, which explicitly expects undo/redo or easy cancellation for common mistakes.

It additionally impacts perceived safety: users become afraid to explore options because any mistake has a heavy cost. Fix: add **Undo**, **Cancel**, or an “Edit pickup time” option; also provide a clear back path without losing the entire flow.

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## Q3) “Confusing language” on a banking transfer screen

**Scenario:** Buttons say “Execute”, “Initiate”, “Proceed”, “Commit” instead of “Send Money”, “Confirm”, “Cancel”.

**Model answer:**

This violates **Match between system and the real world** because the system uses technical/internal terms rather than the user's language and real-world conventions. It also harms learnability and increases cognitive load because users must interpret meaning instead of instantly understanding intent.

From Shneiderman, it reduces **Support internal locus of control**: unclear labels make users feel uncertain and not in control. Fix: replace labels with **user vocabulary** ("Send", "Confirm transfer", "Cancel"), and show a short confirmation summary to increase confidence.

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**Q4) "Same icon does different things" across screens**

**Scenario:** A "bell" icon opens notifications on the home screen, but opens "sound settings" on profile screen.

**Model answer:**

This violates **Consistency and standards** because the same symbol should map to the same action; otherwise users build the wrong mental model. It also violates Shneiderman's **Strive for consistency** (visual + behavior consistency).

Impact: users make errors, hesitate, and feel the app is unpredictable. Fix: keep the bell consistently for notifications, and use a standard "speaker" or "gear" icon for sound/settings.

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**Q5) "Error message without recovery"**

**Scenario:** Form shows: "Error 403: Invalid Request" after clicking Pay. No hint what to do next.

**Model answer:**

This violates Nielsen's **Help users recognize, diagnose, and recover from errors** because the message does not explain the cause in human terms or provide recovery steps. It also violates **Help and documentation** if there's no accessible support path.

From Shneiderman, it fails **Error prevention + simple handling** because it neither prevents the problem nor guides the user through resolution. Fix: replace with actionable text ("Payment failed: card declined / network issue"), preserve user input, and offer "Try again", "Change method", or "Contact support".

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**Q6) "Too many choices" on the first screen**

**Scenario:** Homepage shows 14 equal-weight buttons (Ride, Courier, Food, Grocery, Truck, Rentals...).

**Model answer:**

This violates **Aesthetic and minimalist design** because too much equal-priority content creates clutter that hides what most users need. It also increases memory and attention load, conflicting with Shneiderman's **Reduce short-term memory load** (users must scan/compare many options at once).

Fix: prioritize top 3–4 frequent tasks, group the rest under “More services”, and use progressive disclosure so the UI is simpler and faster for the majority.

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**Q7) “No feedback” after pressing a button**

**Scenario:** User taps “Book Ride”; button doesn’t change, screen stays still for 6 seconds, then suddenly jumps.

**Model answer:**

This violates **Visibility of system status** because users are not kept informed that something is happening. It also violates Shneiderman's **Offer informative feedback**, since actions must have immediate system response (loading, progress, state change).

Impact: users double-tap, create duplicate requests, and lose trust. Fix: show a loading state (“Searching for drivers...”), animate button pressed state, and provide progress feedback.

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**Q8) “Important control looks like plain text”**

**Scenario:** “Delete Account” appears as normal body text at the bottom; many users don’t realize it’s clickable.

**Model answer:**

This violates **Recognition rather than recall** and (more basically) **visibility** because the action is not clearly signified as an interactive control, so users cannot recognize it as a button. In heuristic-evaluation terms, it becomes a discoverability problem: users miss critical actions.

Fix: use standard affordances/signifiers: a clearly styled button/link, spacing, icon (trash), and warning copy. Also add confirmation and undo path to satisfy user control.

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**Q9) “Different back behaviors”**

**Scenario:** On Android, sometimes back arrow goes to previous screen; sometimes it closes the app; sometimes it clears the form.

**Model answer:**

This violates **Consistency and standards** because platform conventions (especially Back

behavior) must be predictable. It also violates **User control and freedom** since unexpected back actions remove user progress and feel like loss of control.

Fix: adopt a single navigation rule: back returns to previous state without destroying data, and provide explicit “Cancel” if you want to discard a flow.

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## Q10) “Forces memorization” for address entry

**Scenario:** User must type full pickup address each time; there are no recent/saved suggestions.

**Model answer:**

This violates **Recognition rather than recall** because the interface forces users to remember and retype details instead of recognizing from saved options. It also violates Shneiderman’s **Reduce short-term memory load**, which expects the UI to support memory with visible choices and chunking.

Fix: show recent locations, saved places (Home/Work), autocomplete, and map pin selection to reduce effort and errors.

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## Q11) “No closure” after completing payment

**Scenario:** After payment, it returns to home silently. No receipt, no confirmation, no summary.

**Model answer:**

This violates Shneiderman’s **Design dialogs to yield closure** because users need a clear “task finished” state with summary and next steps. It also violates **Visibility of system status** because users cannot tell whether payment succeeded or failed.

Fix: show a success confirmation screen with receipt details, send notification/email, and provide a clear path (“View booking”, “Download receipt”).

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## Q12) “Prevents exploration” via irreversible destructive actions

**Scenario:** One tap on “Remove card” instantly deletes it; no confirmation; no undo.

**Model answer:**

This violates **Error prevention** because the design allows a high-impact error with no guardrails. It also violates Shneiderman’s **Error prevention + simple handling** and **Permit easy reversal of actions**, since destructive actions should be confirmable and recoverable.

Fix: add confirmation (“Remove card ending 1234?”), allow undo for a short time, and clearly communicate consequences.

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### Q13) “Help missing” in a complex flow

**Scenario:** Insurance claim screen asks for “Policy rider, nominee endorsement, CNIC back scan” with no explanation or help.

**Model answer:**

This violates **Help and documentation** because when a task is complex or unfamiliar, users need accessible guidance. It also violates **Match with the real world** if jargon is used without translation.

Fix: add inline tooltips, examples (“Where to find this on your policy”), and a help link or chat support so users can complete the task correctly.

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### Q14) “Shortcut missing” for expert users

**Scenario:** Frequent users must go through 6 screens every time to repeat the same monthly bill payment.

**Model answer:**

This violates **Flexibility and efficiency of use** because expert users benefit from shortcuts, personalization, and faster paths. It also conflicts with Shneiderman’s **Cater to universal usability**, which explicitly says support both novice and expert users (e.g., shortcuts for frequent users).

Fix: add “Pay again”, saved billers, templates, and a one-screen confirmation for repeat payments.

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### Q15) “Mobile context ignored”

**Scenario:** A mobile form logs users out after 15 seconds of inactivity; no draft saving; no warning. Many users get interrupted mid-entry.

**Model answer:**

This violates **User control and freedom** because users lose work due to interruption without an easy recovery path. It also violates the mobile reality that users are **interruptible and easily distracted** (calls/notifications/real-world interruptions), so the system must be tolerant of pauses.

Fix: autosave drafts, show timeout warnings, allow session resume, and preserve user progress after interruptions.

## Q16) “Greyed out button with no explanation”

### Scenario:

The “Submit” button is disabled (greyed out). There is no indication why. Users keep tapping it.

### Model Answer:

This violates **Visibility of system status** because the system does not communicate *why* an action is unavailable. It also violates **Help users recognize, diagnose, and recover from errors**, since users cannot identify what condition is missing.

From Shneiderman’s rules, it fails **Offer informative feedback**, because disabling alone is not sufficient feedback without explanation.

Fix: show inline hints (“Please upload CNIC front & back”), highlight missing fields, or show a tooltip on the disabled button.

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## Q17) “Color-only error indication”

### Scenario:

Invalid form fields turn red, but there is no icon, text, or message.

### Model Answer:

This violates **Accessibility** because color alone is used to convey critical information, which excludes color-blind users. It also violates **Help users recognize, diagnose, and recover from errors**, as users may not understand what is wrong or how to fix it.

Cognitively, it increases perceptual ambiguity and slows correction. Fix: combine color with **icons, labels, and error text**, ensuring redundant cues.

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## Q18) “Ambiguous confirmation message”

### Scenario:

After pressing “Confirm”, a popup appears saying: “Done.”

### Model Answer:

This violates **Visibility of system status** because “Done” does not clearly communicate *what* action succeeded. It also violates Shneiderman’s **Design dialogs to yield closure**, which requires meaningful completion feedback tied to the task.

Users cannot evaluate system state (Gulf of Evaluation), increasing uncertainty. Fix: replace with explicit feedback such as “Ride booked successfully – Driver arriving in 4 minutes.”

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## Q19) “Mismatch with real-world process”

### Scenario:

A vending machine app asks users to select item *before* inserting payment, without guidance.

### Model Answer:

This violates **Match between system and the real world** because it conflicts with users' established mental models of vending machines (pay → select). When strong conventions are broken, the system must compensate with clear guidance.

The result is confusion and the belief that the system is broken. Fix: either follow the conventional flow or provide strong visual instructions and feedback to teach the new interaction.

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## Q20) “Overloaded settings screen”

### Scenario:

Settings page shows 30 toggles in one long scroll, all same size and importance.

### Model Answer:

This violates **Aesthetic and minimalist design** because it overwhelms users with too much undifferentiated information. It also violates **Reduce short-term memory load**, as users must remember what each toggle does while scrolling.

Fix: group related settings, use headings, progressive disclosure, and prioritize frequently used options.

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## Q21) “Unexpected automatic action”

### Scenario:

As soon as a user opens the app, it automatically places an order using the last used address.

### Model Answer:

This violates **User control and freedom** because the system initiates a high-impact action without explicit user confirmation. It also violates **Support internal locus of control**, making users feel the system is acting on its own.

Fix: always require explicit confirmation for irreversible or costly actions, even if defaults are suggested.

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## Q22) “Poor mapping in controls”

### Scenario:

Volume and brightness sliders are stacked vertically but affect opposite sides of the screen.

### Model Answer:

This violates **Mapping**, as the spatial relationship between controls and effects is illogical. Users must think instead of intuitively understanding the relationship.

Poor mapping increases cognitive load and slows interaction. Fix: align controls spatially with their effects or use clear icons and labels to reinforce mapping.

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## Q23) “Hidden destructive action”

### Scenario:

Swipe left on a list item instantly deletes it. No icon, no warning.

### Model Answer:

This violates **Error prevention** because a common gesture performs a destructive action without safeguards. It also violates **Permit easy reversal of actions**, since there is no undo.

Fix: add confirmation, visual affordances (trash icon), or temporary undo (“Item deleted – Undo”).

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## Q24) “Inconsistent terminology”

### Scenario:

Same feature is called “Trips” on one screen and “Rides” on another.

### Model Answer:

This violates **Consistency and standards**, as inconsistent terminology breaks the user’s mental model. It increases learning effort and causes hesitation.

Fix: standardize language across the system and align with user vocabulary.

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## Q25) “Keyboard shortcut overload”

### Scenario:

Desktop app assigns Ctrl+S, Ctrl+Shift+S, Ctrl+Alt+S to different unrelated functions.

**Model Answer:**

This violates **Consistency and standards**, especially external consistency, because it conflicts with widely learned shortcuts. It also increases cognitive burden by forcing memorization.

Fix: follow platform conventions and avoid overloading familiar shortcuts.

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## Q26) “No progress indicator”

**Scenario:**

File upload takes 20 seconds. Screen is static.

**Model Answer:**

This violates **Visibility of system status** because users are not informed about ongoing processes. It leads to repeated clicks and frustration.

Fix: show progress bars, percentages, or animated indicators to reassure users.

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## Q27) “Multiple identical buttons”

**Scenario:**

Three buttons on a screen all say “Continue” but lead to different paths.

**Model Answer:**

This violates **Recognition rather than recall** because users cannot distinguish actions based on labels alone. It also violates **Consistency**, as identical labels should produce identical outcomes.

Fix: use descriptive labels (“Continue to Payment”, “Continue as Guest”).

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## Q28) “Auto-logout without warning”

**Scenario:**

User is logged out due to inactivity and all entered data is lost.

**Model Answer:**

This violates **User control and freedom** because users cannot recover from interruption. It also violates mobile context assumptions where interruptions are common.

Fix: autosave drafts, warn before logout, and allow session recovery.

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## Q29) “Unclear iconography”

### Scenario:

An abstract icon is used for “Report Issue” with no label.

### Model Answer:

This violates **Recognition rather than recall**, as users must guess icon meaning. It also weakens affordances and discoverability.

Fix: add text labels or tooltips, especially for critical actions.

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## Q30) “Too much help text upfront”

### Scenario:

On first screen, user is shown a full page of instructions before doing anything.

### Model Answer:

This violates **Aesthetic and minimalist design**, as unnecessary information clutters the interface. It also overwhelms attention and discourages exploration.

Fix: progressive disclosure—show help when needed, not all at once.

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## Q31) “No differentiation between primary & secondary actions”

### Scenario:

“Delete Account” and “Save Changes” look identical.

### Model Answer:

This violates **Error prevention**, as dangerous actions should be visually distinguished. It also violates **Aesthetic and minimalist design** by failing to establish visual hierarchy.

Fix: color, placement, spacing, and confirmation dialogs for destructive actions.

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## Q32) “Non-standard gesture with no signifier”

### Scenario:

User must long-press an image to access options; nothing suggests this.

**Model Answer:**

This violates **Visibility** and **Recognition rather than recall** because the gesture is hidden and undocumented. Users cannot discover functionality without prior knowledge.

Fix: add visual signifiers, hints, or alternative visible controls.

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### Q33) “System blame in error messages”

**Scenario:**

Error says: “User input invalid.”

**Model Answer:**

This violates **Help users recognize, diagnose, and recover from errors**, because it blames the user without explaining the issue. It also creates negative emotional response.

Fix: polite, specific messages (“Phone number must be 11 digits”).

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### Q34) “Too many steps for a simple task”

**Scenario:**

Turning on notifications requires navigating through 5 screens.

**Model Answer:**

This violates **Flexibility and efficiency of use**, especially for frequent users. It increases friction and discourages engagement.

Fix: shortcuts, quick toggles, and direct access to frequent actions.

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### Q35) “No way to review before final action”

**Scenario:**

User enters payment details and payment happens immediately.

**Model Answer:**

This violates **Error prevention** and **Design dialogs to yield closure**, because users need a final review step before irreversible actions.

Fix: add a confirmation summary screen with editable fields.

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## Q36) “Inconsistent visual hierarchy”

### Scenario:

Important warnings look smaller than promotional banners.

### Model Answer:

This violates **Visibility** and **Aesthetic and minimalist design**, as visual priority does not match importance.

Fix: emphasize warnings with size, contrast, and placement.

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## Q37) “Forced sign-up before exploration”

### Scenario:

User must create account before even browsing content.

### Model Answer:

This violates **User control and freedom**, limiting exploration and increasing commitment too early.

Fix: allow guest mode, preview content, then request sign-up when value is clear.

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## Q38) “No consistency with platform conventions”

### Scenario:

iOS app uses Android-style navigation patterns.

### Model Answer:

This violates **Consistency and standards**, particularly platform standards users already know.

Fix: follow Apple HIG or Android Material guidelines accordingly.

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## Q39) “Misleading affordance”

### Scenario:

Text looks like a link but is not clickable.

### Model Answer:

This violates **Affordances** because visual cues suggest an action that does not exist.

Fix: align appearance with behavior; remove misleading styling.

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## Q40) “No onboarding for novel interaction”

### Scenario:

App introduces gesture-based navigation with no explanation.

### Model Answer:

This violates **Help and documentation** and increases the **Gulf of Execution**, as users don't know what actions are possible.

Fix: lightweight onboarding, coach marks, or contextual hints.